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Transparency in transnational governance: The determinants of information disclosure of voluntary sustainability programs

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Abstract

The rise of "new" transnational governance has intensified debates about a lack of accountability in global politics. Reviewing the mechanisms through which transparency can foster accountability beyond the state, this article explores the determinants of information disclosure in the field of transnational sustainability governance. Examining the institutional design of 113 voluntary sustainability programs, we find a positive correlation between the involvement of public actors and information disclosure. In contrast, the role of civil society is more ambiguous. There is no statistical support for arguments linking non-governmental organization participation to increased transparency. At the same time, our analysis reveals a robust correlation between civil society-led metagovernance and information disclosure. Moreover, we find that crowding has a negative effect on transparency, whereas normative peer pressures have no influence. At a broader level, the analysis reveals a lack of "deep transparency" among transnational sustainability governors. This limits the scope for transparency-induced accountability in this policy domain.

Keywords: accountability, institutional design, sustainability, transnational governance, transparency.

1. Introduction

When it comes to the governance of transnational production, "the state is far from the only game in town" (Abbott & Snidal 2009a, p. 87). This is particularly true of the field of global sustainability politics where private and hybrid governance arrangements have proliferated in recent years. Prominent examples are the Forest Stewardship Council (FSC) and the Rainforest Alliance. Created by non-governmental organizations (NGOs) and firms (sometimes with the involvement of governments), these voluntary sustainability programs (VSPs) set standards for transnational production, operate verification systems, and feature quasi-judicial conflict resolution mechanisms. In this and other policy areas, they have taken over a wide range of governance functions that used to be the prerogative of states and international organizations (IOs).

This shift from state-centered to polycentric governance has intensified debates about a lack of accountability in global politics (Grant & Keohane 2005; Bäckstrand 2006; Buchanan & Keohane 2006; Black 2008; Biermann & Gupta 2011; Kramarz & Park 2016; Bäckstrand *et al.* 2018). Of particular concern is the rapidly expanding realm of transnational private governance, with its many actors and diffuse authority structures. Here, traditional notions of accountability no longer apply (Dingwerth 2007). In search of alternatives, transparency is often hailed as a possible solution. Transparency, it is argued, can foster accountability in transnational governance by

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enabling mechanisms of "soft enforcement," such as market pressures, public scrutiny, and self-reflection (Florini 2003; Hale 2008). Others, however, are more skeptical. They critically scrutinize the ability of transparency to empower accounting actors in global governance. Transparency may indeed not be a silver bullet, but even these critical scholars continue to view it as a positive force (Gupta 2010; Mol 2010; Gupta & Mason 2014).

While the current discussion is mainly centered on the effects of transparency, less is known about its sources. If transparency matters (and the broader debate seems to suggest just that), we need to better understand its origins. Several studies have examined the sources of transparency in the context of IOs (Mitchell 1998; Grigorescu 2007); however, in the realm of transnational governance, these relationships remain underresearched.[1] Given these gaps in our understanding of the issue, this article explores the determinants of information disclosure in global sustainability politics – a policy domain that has seen a "Cambrian explosion" of new transnational governance in recent years (Abbott 2012).

To shed light on the issue, we analyze a dataset of 113 VSPs and develop a new measure of transparency. With a focus on VSP's operations, we describe their transparency practices in four areas: decisionmaking, standard setting, verification, and dispute settlement. Our results reveal that many VSPs are "shallow" in their transparency – that is, they disclose little about the ways in which they work in practice. We argue that this limits the scope for transparency-induced accountability in this governance domain. At the same time, we observe significant interprogram variation. To investigate these differences, we review the broader literature on transparency, combining insights from studies on domestic politics, IOs, and transnational governance in order to develop six hypotheses about the internal and external determinants of information disclosure, which we test in a multivariate analysis. The analysis reveals a positive correlation between the involvement of public actors in VSPs' central governance bodies and information disclosure. The role of civil society is more ambiguous. There is no statistical support for the widely held belief that direct NGO participation increases transparency levels. However, the analysis reveals a robust correlation between civil society-led metagovernance and information disclosure. Moreover, we find that crowding has a negative influence on transparency, whereas normative peer pressure has no significant effect. We interpret these findings in light of existing studies on transnational governance, complementing the mostly qualitative literature on the subject.[2]

The remainder of this article is organized in seven sections. Section 2 revisits the debate on accountability and transparency in transnational governance. Section 3 introduces the concept of transparency and presents the results of the descriptive analysis. In sections 4 and 5, we introduce and operationalize our hypotheses and discuss our empirical model. Section 6 and 7 present the results of the multivariate analysis and interpret our main findings. A final section concludes.

2. Accountability and transparency in transnational governance

At the beginning of the 1990s, the end of the Cold War ushered in an era of transnational governance, opening up space for non-state actors to play a more salient role in international affairs (Rosenau 1992). While the growth of formal IOs has slowed, the population of private and hybrid governance arrangements has increased exponentially (Abbott & Snidal 2009a; Abbott *et al.* 2016). This is particularly true in the field of sustainability governance where growth of the number of private and hybrid governance arrangements has accelerated over the last 20 years (Abbott 2012). One important first-mover program was the FSC (Cashore *et al.* 2004). Created in 1993 by a coalition of environmental NGOs and firms, the program sets standards for sustainable forestry management and operates a certification program with global reach. Since then, a great variety of VSPs have been created in a wide range of industry sectors. Other examples include well-known programs like the Roundtable on Sustainable Palm Oil and Rainforest Alliance. However, there are also many less visible VSPs, such as Fair Flowers Fair Plants or the ProTerra Foundation. Figure 1 illustrates this trend over time based on our research sample (see discussion).

Among scholars of international relations (IR), the proliferation of new modes of governance has intensified debates about the prospects and limits of accountability beyond the state (Grant & Keohane 2005; Bäckstrand 2006; Dingwerth 2007; Black 2008; Hale 2008; Biermann & Gupta 2011; Kramarz & Park 2016; Bäckstrand *et al.* 2018). One important concern is that global governance institutions are too distant and detached from citizens. The result is a growing accountability deficit as more and more decisionmaking authority is transferred from the national to international and transnational levels (Dingwerth 2007).

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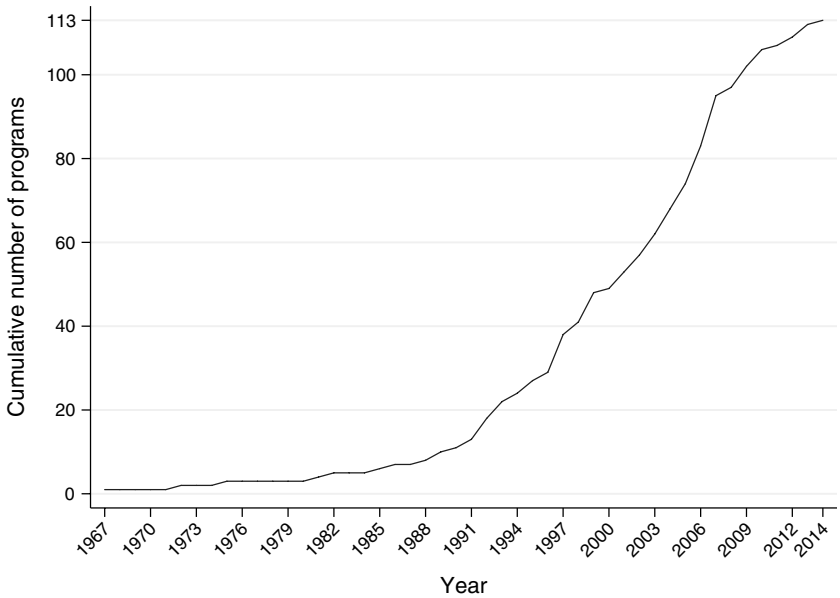

Figure 1 Proliferation of voluntary sustainability programs over time.

The problem, in a nutshell, is that at the national level, an important source of democratic legitimacy is what Grant and Keohane (2005, p. 29) call principal–agent accountability. In this model, the principal (people) has the right and ability to hold its agent (government) to account, to judge whether it has fulfilled its responsibilities, and to impose sanctions if these responsibilities have not been met. In democratically constituted states, the main mechanism to achieve this is through periodic general elections. However, in governance beyond the state, principal–agent accountability of this type runs into difficulties. In the realm of transnational governance, the model does not work. Here, no principal (or global demos) exists, governors are typically self-selected, and no electoral mechanism is available to hold them to account (Dryzek 2000; Dingwerth 2007).

In this context, much hope has been put on the concept of transparency (Florini 2003; Hale 2008). Hale noted"[i]f 'democracy deficit' is the catchphrase for global governance's problem, 'transparency' is its buzzword solution" (2008, p. 73). But what role can transparency play in mitigating the problem? An answer to this question requires unpacking the concepts of accountability and transparency and conducting a discussion on how they are related.

According to Schedler, the concept of accountability has two components: answerability – "the right to receive information and the corresponding obligation to release details" – and enforcement – "the idea that accounting actors do not just 'call into question' but also eventually punish improper behavior" (1999, p. 13). As summarized by Hale (2008), A is thus accountable to B if B can (i) know A's behavior; and (ii) exert pressure on A to influence that behavior. Regarding the first component of accountability, the importance of transparency for establishing accountability is clear. Without reliable information, answerability is impossible. However, the role of transparency in enforcement is less obvious and also more controversial.

Optimists argue, "transparency is providing new opportunities both to enforce rules and standards and to hold accountable those who purport to act in the public interest" (Florini 2003, p. 196). But how does this work in practice? In *Full Disclosure: The Perils and Promise of Transparency,* Fung et al. (2008) describe a "transparency action cycle" in which information disclosure triggers constructive behavioral change. Their causal model proceeds in four stages: (i) a discloser (e.g. a company) discloses information (e.g. pollution data) that is relevant and salient to users (e.g. consumers); (ii) users act in response to this information; (iii) the discloser is sensitive to users' actions; and (iv) responds constructively. From an enforcement perspective, the interesting question is: What mechanisms of behavioral change (stages 2–4) does transparency trigger? And how effective are they?

Reviewing the literature on transnational governance, Hale (2008, pp. 76–87) identifies three main mechanisms that accounting actors can use to hold targeted institutions accountable. First, market pressure plays an important role. For example, consumers, investors, and NGOs can respond to information disclosure by changing

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their consumption and investment decisions or by launching corporate shame campaigns. These actions can unfold a coercive force if they threaten to have material consequences for the target actor. Second, there are methods of "soft enforcement" through public discourse. Based on Habermasian discourse theory, the argument runs that transparency creates pressure to tell the truth, as it makes it easier to expose lies through the "forceless force of the better argument." This can discourage rent seeking and other self-serving behavior. Finally, transparency can facilitate behavioral change through enabling self-reflection. In this regard, information disclosure can reveal discrepancies between an actor's internalized norms and its actual behavior and a desire to correct the mismatch.

However, these mechanisms have their limitations, and scholars have expressed doubts about transparency and its ability to empower and enforce in transnational governance. For example, while acknowledging the importance of information for all forms of accountability, Grant and Keohane state "[w]ithout standards and sanctions (…) accountability that is both effective and widely viewed as legitimate will remain elusive" (2005, pp. 39–40). In addition, several scholars that have explored the role of transparency in global environmental governance have expressed skepticism about its ability to truly empower accounting actors (Gupta 2010; Mol 2010; Gupta & Mason 2014). Thus, as observed by Fox (2007), the relationship between transparency and accountability remains uncertain. Hard accountability that includes sanctions might indeed remain elusive in transnational governance. But like Hale, Fox believes that transparency can lead to softer forms of accountability.

The upshot of this discussion is that transparency is not a panacea to the accountability deficit of transnational governance (Grant & Keohane 2005). However, the disclosure of salient information may enable soft enforcement through market pressure, public discourse, and self-reflection. What seems significant in this context is the "depth" of information that is being disclosed. By this we mean the degree to which the disclosed information allows insights into the actual workings of an institution, not just its formal procedures. As noted by Fox (2007), transparency that reveals little about the ways in which an institution works in practice will not do the job. But to be clear, we are not assuming that transparency is a sufficient condition for generating accountability beyond the state. However, we follow Hale and others who argue that transparency is necessary for holding transnational governors to account.

3. Conceptualizing and measuring transparency

Given the centrality of transparency to the debate on accountability in transnational governance, our goal is to investigate it empirically. A look at the existing scholarship reveals that there are many different ways to study the phenomenon. For example, in the realm of domestic politics, there is a large body of literature focusing on the adoption and design of so-called freedom of information policies (e.g. Berliner 2014). In a similar way, scholars of IR have begun to analyze and compare the transparency policies of global governance institutions (Grigorescu 2003; Donaldson & Kingsbury 2013). Studying formal transparency polices has clear advantages. They are relatively easy to analyze and compare. However, there are also drawbacks. Most importantly, formal policies and actual practices often differ significantly (Florini 2002). In the context of domestic governance, accounting actors can resort to legal mechanisms to enforce transparency policies. However, in the transnational realm this is not possible.

Therefore, instead of formal policies, we focus on information disclosure practices. This perspective highlights the relational character of transparency. Studying it empirically requires specification of *what* is being disclosed, *by whom*, and *for whom* (Grigorescu 2007; Gupta & Mason 2014, p. 5). Regarding the "what" and by "whom" questions, we focus our analysis on the transparency of VSPs about their operations. We construct our measure of transparency based on what we identify as their main activities in the regulatory process, namely: decisionmaking, standard setting, verification, and dispute settlement. For each of these areas, two variables, representing different degrees of transparency, are selected from a database (see discussion). We include measures for "shallow transparency" – disclosure of information about formal procedures – and "deep transparency" – disclosure of information about actual processes (see Table 1).

In addition to identifying the discloser and the object of transparency, it is also important to define the target audience – that is, the group of actors to whom information is being disclosed (Grigorescu 2007, pp. 626–629). We focus on disclosure to the general public, which is the most unrestricted form of information disclosure. More

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Table 1 Measures of transparency

Area of operation/ degree of transparency	Shallow	Deep
Decisionmaking	Disclosure of information about the composition of the main governing body	Disclosure of meeting minutes of the main governing body
Standard-setting	Disclosure of documents about the content of standards	Disclosure of information about the standard-setting process (e.g. comments from stakeholder consultations)
Verification	Disclosure of information about certificate holders (e.g. name of company, validity of certificate)	Disclosure of information about certification decisions (e.g. audit reports)
Dispute settlement	Disclosure of information about dispute resolution policies	Disclosure of information about dispute resolution decisions (e.g. documentation about individual cases)

precisely, we focus on the open disclosure of information about decisionmaking, standard setting, verification, and dispute settlement on VSPs websites. Overall, this leaves us with a clearly delineated concept of transparency for our investigation.

To measure transparency empirically, we source data from the Standards Map of the International Trade Centre (ITC).[3] Launched in 2011, the Standards Map database (SMD) is an inventory of VSPs in the field of trade and production. In October 2018, the SMD included 247 VSPs operating in more than 120 product groups and 180 countries. It is one of the most comprehensive resources on VSPs and contains information about their standards, geographic scope, and organizational processes (including detailed information about their disclosure practices, as listed in Table 1).[4]

According to the ITC's data collection protocols, the VSPs included in the database all satisfy the following minimum criteria: existence of a published set of criteria and indicators; existence of an implementation system; and coverage of at least one sustainability area (environment, social, economic and management, quality management, or ethics and integrity). In addition, the SMD includes transnational and domestic programs as well as VSPs with varying sponsorship arrangements (e.g. industry, civil society, and multistakeholder).[5]

One important limitation of the SMD is its weak coverage of certain types of VSPs, particularly firm-level programs. Several company codes of conduct are listed; however, the SMD's coverage in this area cannot be considered representative. A second limitation of the SMD is its "snapshot character." As updates are performed at annual intervals, the information it contains reflects the state of affairs at the latest update. This limits the possibility to use the SMD to investigate trends and dynamics over time. Thus, we are unable to examine arguments by, for example, Gupta and Mason (2014, p. 15) who hypothesize that processes of democratization and marketization have driven the uptake of transparency in global sustainability governance.[6]

To gain access to the full SMD and to prepare and work with the data, we undertook three field trips to ITC headquarters in Geneva in November 2015, February 2017, and October 2017. The dataset analyzed in this study is a subsample of 113 VSPs taken from the SMD. The following selection criteria were applied in the creation of the sample: (i) the program is operational; (ii) it has a discernible governance structure; (iii) it is transnational (i.e. it operates in more than one country); (iv) it is not a firm-level code of conduct; (v) it develops environmental and/or social standards for global supply chains (no purely technical, food safety, or quality standards); and (vi) in the selection of our sample, we took into account that some VSPs are listed with multiple standards in the SMD.

Presenting the results of a first-cut analysis, Figure 2 shows that transparency levels vary significantly in this group of programs. The vast difference between shallow and deep transparency is particularly striking. The latter is much less frequent, and this is true for all four areas of operation. A common sense explanation would be that disclosing sensitive information about actual processes (as opposed to formal procedures) is associated with higher costs for the discloser – for example, by enabling more thorough public scrutiny. In addition, we observe variation across areas of operation. In this regard, Figure 2 reveals that programs are most transparent in the area

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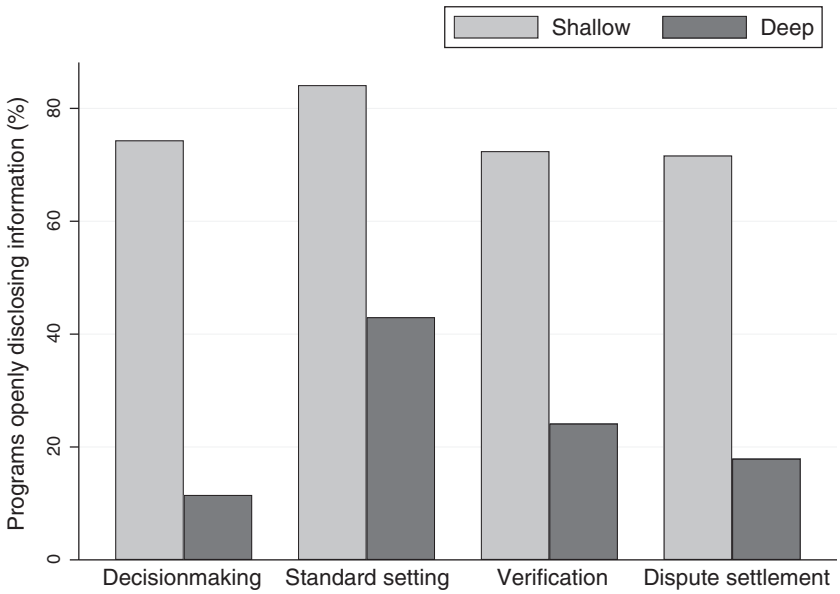

Figure 2 Shallow and deep transparency across areas of operation.

of standard setting. In contrast, transparency levels are much lower in the areas of decisionmaking, verification, and dispute settlement. Here, the vast majority of VSPs (> 75%) do not disclose information at a deep level of transparency. In addition to finding differences in transparency levels across areas of operation, the data revealed that there is a large group of 54 VSPs that do not disclose deep information at all. At the same time, we identified only two programs that engage in deep transparency in all four areas of operation.

Against this background, we hypothesize that in many cases transparency levels are insufficient to enable the kind of "soft enforcement" described by Fox (2007), Hale (2008), and others. While we did not analyze these mechanisms directly, our descriptive results suggest that the most important scope condition of the transparency action cycle suggested by Fung *et al.* (2008) is currently not met by a large group of VSPs. Recall the first stage in the model assumes that a discloser discloses information that is relevant and salient to users. In a second stage, they then act in response to this information and the transparency action cycle starts to spin. However, as shown above, many VSPs do not to disclose information that truly reveals the ways in which they work in practice. Thus, achieving accountability through transparency becomes unlikely in these cases.

At the same time, however, we also observe significant interprogram variation (which we will examine in more detail). In preparation for the empirical analysis, we use a simple strategy to aggregate the available information and distill a synthetic, comprehensive transparency index that takes a unique value for each VSP. Therefore, we compute weighted averages of the disclosure dummies described above, across both the different areas of operations and the two levels of transparency. In the absence of a refined theory on the issue, we weigh areas of operations symmetrically, assuming that they are all equally important in determining the overall proxy of transparency. In contrast, we allow for different weighting schemes of the two levels of transparency, following the common sense intuition that deep transparency is more important than shallow transparency.

In order to provide a formal definition of the index, we first introduce some simple notation. The area of operation (varying between decisionmaking, standard setting, verification, and dispute settlement) is indexed with c while the level of transparency (either shallow or deep) is denoted with l. For each program s, area of operation c, and level l, we define a dummy variable $t_{s;\,c,\,l}$ that takes a value of 1 if the program s publicly discloses information at level l about area of operation c (and 0 otherwise). The transparency indicator is then constructed in three steps. First, we take a weighted sum of $t_{s;\,c,\,l}$ across areas and levels. Second, we divide the result by the number of areas of operation in which a program is active. Formally:

$$TI_s = \frac{1}{\sum_c \mathbb{I}_s(c)} \sum_l \sum_c w_l \times t_{s;c,l} \times \mathbb{I}_s(c) \tag{1}$$

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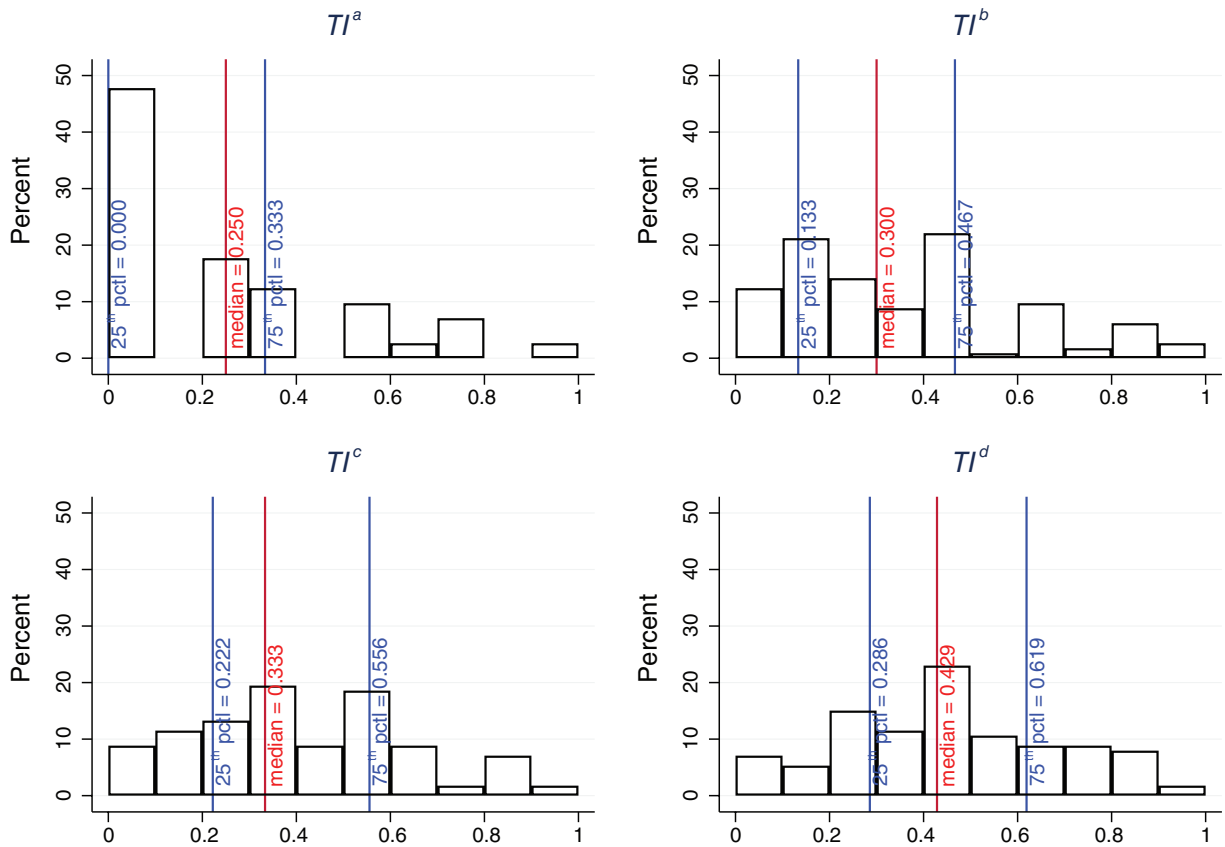

Figure 3 Distribution of the transparency indices (percent of programs).

where $\mathbb{I}_s(c)$ is an indicator that takes a value of 1 if program s is active in the area of operation c. w_l is a weight that applies to level l of transparency. Third, we normalize TI in order to have it vary between 0 and 1 within our sample.

To increase the robustness of the statistical analysis, we adopt four weighting schemes, each of them defining a separate transparency index. This reduces the risk that our results are driven by the specific choices we make in the construction of the index. The first scheme assigns a value of 1 to the deep level and 0 to the shallow one. This version of the index focuses on variation in deep transparency only. We denote the resulting transparency index as TI^a. The second index keeps the 1 unit weight for the deep level, while assigning a weight of 0.25 to the shallow one. Under this scheme disclosure of shallow information does contribute to the measurement of transparency, counting one fourth of its deep-level counterpart. The corresponding index is denoted as TI^b. The third and fourth indices, TI^c and TI^d, further increase the relative importance of shallow transparency by assigning the shallow level weights of 0.5 and 0.75, respectively (while keeping the deep weight at a value of 1).[7] Figure 3 shows the distribution of the four transparency indices for our sample.

As can be seen in Figure 3, the distribution of TI^a reflects the lack of deep transparency identified above. It is clearly skewed to the left-hand side, with a median of only 0.250. Moving to TI^b, observations shift toward the right-hand side of the support, reflecting the higher scores of the index that now assigns a positive weight to shallow transparency. This pattern is amplified with TI^c and even further under TI^d, whose sample median is equal to 0.429 and 25 percent of the observations score a transparency index value higher than 0.619. However, while the structure of the distribution varies across the four versions of our index, the general pattern remains the same: few VSPs are highly transparent. This can be seen from the right tail of the four distributions, which is always relatively thin.

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4. Explaining transparency: Causal mechanisms and hypotheses

Scholarly interest in the role of transparency in transnational governance is growing (Dingwerth & Pattberg 2009; Auld & Gulbrandsen 2010; Overdevest 2010; Gulbrandsen & Auld 2016). A comprehensive analysis of its determinants, however, is still missing. A notable exception is a study by van der Ven (2015), which includes a measure of transparency as part of a broader index of best practice compliance in transnational sustainability governance. In general, however, we still know little about these relationships. To address this gap, this section is grounded in a review of the relevant transnational governance literature, combining it with insights from research on IOs and domestic politics.

The discussion of arguments about transparency is organized around internal and external determinants. The internal determinants are factors that are located at the program-level – that is, its institutional design. In total, three internal determinants are considered: involvement of NGOs, involvement of public actors, and the level of stringency of a program. External determinants are factors located in the institutional environment of a VSP – that is, outside its organizational boundaries. On this dimension another three factors are discussed: crowding, norm diffusion, and metagovernance.

4.1. Internal determinants

4.1.1. Involvement of non-governmental organizations

In IR and related disciplines, NGOs are often portrayed as norm entrepreneurs, promoting democracy, human rights, and environmental protection in global politics (Keck & Sikkink 1998; Risse *et al.* 1999). In the literature on IOs, their integration into policymaking processes has been described as a way to strengthen participation, accountability, and transparency in global governance (Scholte 2011; Tallberg *et al.* 2014). In a similar way, students of transnational governance praise the benefits of multistakeholder initiatives (Cashore *et al.* 2004; Dingwerth 2007; Abbott & Snidal 2009a). Cashore *et al.* described them as "one of the most innovative and startling institutional designs of the past 50 years" (2004, p. 298). Several authors have also drawn a direct connection between the inclusion of NGOs and transparency (Florini 2002; Grigorescu 2007; Gulbrandsen 2008; van der Ven 2015). On the one hand, they describe a "normative mechanism." For example, van der Ven (2015, p. 6) expects that deep NGO involvement in transnational governance will lead to increased attention to best practice out of a desire to serve public ends. On the other hand, scholars have advanced a functionalist explanation. The assumed mechanism is that greater participation from NGOs in governance creates additional demand for information from their constituencies (Welch 2012). Against this background, the following hypothesis is derived:

Hypothesis 1:
VSPs that involve NGOs in their central decisionmaking body are more transparent than those with no NGO involvement.

4.1.2. Involvement of public actors

Connected to the previous discussion about NGO involvement, a second argument concerns the role of public actors (e.g. representatives from state agencies and ministries) in transnational governance. While the emergence of transnational governance institutions has often been analyzed separately from "old" state-led governance (Pattberg 2005), there is growing recognition that public actors play an important role in these processes (Abbott & Snidal 2009a; Gulbrandsen 2014). Some authors see this role in a positive light. For example, Abbott and Snidal argue that greater involvement by public actors could promote "substantive principles and procedures derived from public law to reinforce transparency and accountability, enhancing the legitimacy of private schemes" (2009b, pp. 558). A possible mechanism is the norm entrepreneurship of public actors. Similar to the argument made about NGOs, the assumption is that public officials – at least those from democratically constituted states – believe in the appropriateness of transparency norms and therefore support rules allowing for the open disclosure of information (Grigorescu 2007, pp. 632–633). This leads to the second hypothesis:

Hypothesis 2:
VSPs that involve public actors in their central decisionmaking body are more transparent than those with no public actor involvement.

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4.1.3. Level of stringency

A third internal determinant can be derived from the literature on voluntary environmental programs (Prakash & Potoski 2007; Potoski & Prakash 2009). From this perspective, VSPs are conceptualized as clubs that firms can join to signal their superior sustainability performance to relevant external audiences (e.g. consumers, NGOs, or regulators). For club theorists, the main incentive for firms to do this is to gain branding benefits (Prakash & Potoski 2007). These benefits crucially depend on the level of stringency of a program – that is, the design of its standards and monitoring and enforcement procedures. The reason is that, everything else being equal, more stringent programs create higher positive externalities (e.g. a reduction of environmental impacts). This strengthens the program's reputation, thus affecting the branding benefits received by individual members. In this regard, Prakash and Potoski explain how a "standards' stringency serves as a proxy signal for the level of externalities members generate (per capita) and therefore affects the branding benefits members can expect to receive from stakeholders" (2007, p. 7). However, this logic only works if programs make this information openly available. Otherwise, no (or only weak) signals are sent. Following from this, we hypothesize that stringent programs have a particularly strong incentive to disclose information about their operations. This would send the strongest possible signal, thus maximizing the branding benefits for their members.

Hypothesis 3:
Stringent VSPs are more transparent about their operations than less stringent ones.

4.2. External determinants

4.2.1. Crowding

The decentralized evolution of transnational regulatory regimes has created considerable overlap in issue and industry coverage. This has led to competition between VSPs (Overdevest 2010; Fransen 2011; Schleifer 2013; Eberlein *et al.* 2014; Overdevest & Zeitlin 2014). However, the effects are not yet fully understood. Certain studies have shown that a "ratcheting-up effect" is possible. In this regard, investigating the interactions between VSPs in the forestry sector, Overdevest (2010) describes how the Program for the Endorsement of Forest Certification (PEFC) upgraded its standards and procedures, including its transparency practices, in response to regulatory competition with the NGO-backed FSC. Drawing on the business studies literature, she describes a public benchmarking mechanism – a process of comparing practices between competing programs in order to achieve improvements. As shown in the case of the forestry sector, this can have a "ratcheting-up effect" if program managers and external stakeholders evaluate such practices positively. However, the kind of public benchmarking that occurred between the PEFC and FSC appears to be the exception rather than the rule. Several other studies on VSP interactions describe "race-to-the-bottom" dynamics between competing programs (Fransen 2011; Schleifer 2013; Marx & Wouters 2014). Based on the literature on organizational ecology, we hypothesize that such downward pressures are most likely to occur in "crowded" environments in which high numbers of programs overlap with one another, resulting in resource competition between VSPs (Hannan & Freeman 1989).

Hypothesis 4:
VSPs that operate in highly crowded environments are less transparent.

4.2.2. Peer pressure

An important argument in the literature on norm diffusion is that processes of norm adoption are interdependent (DiMaggio & Powell 1991; Strang 1991). In this regard, Strang describes how the "prior adoption of a trait or practice in a population alters the probability of adoption for remaining non-adopters" (1991, p. 325). The mechanism works through the "logic of appropriateness" (March & Olsen 1998). As a norm diffuses in a population of organizations, adoption becomes the "appropriate" thing to do and non-adopters risk challenges to their legitimacy or even their survival. This peer pressure mechanism has been described in several studies examining transparency practices at the domestic, international, and transnational level (Dingwerth & Pattberg 2009; Donaldson & Kingsbury 2013; Berliner 2014). For example, Donaldson and Kingsbury (2013) argue that global

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governance institutions become more receptive to transparency norms if their peer institutions have adopted such policies. Against this background, the following hypothesis is derived:

Hypothesis 5:
VSPs are more transparent if other programs in their institutional environment have adopted high transparency standards.

4.2.3. Metagovernance

Transnational regulatory fields are increasingly structured by metagovernance organizations. These are organizations that create principles and criteria of good practice for standard-setting bodies worldwide. Important metagovernance organizations include the Global Social Compliance Programme, the International Standardization Organization, and the International Social and Environmental Accreditation and Labelling (ISEAL) Alliance. Founded by a group of NGO-backed VSPs in 2002, ISEAL is commonly considered to be the focal metagovernor in the field of transnational sustainability governance (Dingwerth & Pattberg 2009; Glasbergen 2011; Loconto & Fouilleux 2014). All VSPs aiming to become a member of ISEAL need to go through a benchmarking and accreditation procedure in which their compliance with best practices, such as ISEAL's standard-setting code, is verified. Currently, ISEAL counts 23 members.[8] Loconto and Fouilleux (2014) describe how ISEAL acts as an important institutional entrepreneur in the field of sustainability governance as it promotes credibility principles, including transparency norms. In a similar vein, Dingwerth and Pattberg (2009) argue that ISEAL exercises normative pressure on VSPs. However, other studies have shown that the resulting "isomorphic effect" is limited (Schleifer 2019). Still, most of the above-cited studies would expect a positive relationship between participation in metagovernance organizations and transparency.

Hypothesis 6:
VSPs that are members of metagovernance organizations are more transparent then those that are not.

5. The determinants of transparency: A multivariate analysis

In this section, we describe the operationalization of our independent variables and the statistical model we use, followed by the results of our multivariate analysis.

5.1. Operationalization

We begin the operationalization with our *internal* regressors. To operationalize the involvement of NGOs (H1) and public actors (H2) we use two dummy variables taking a value of 1 if NGOs and public actors, respectively, are involved in the central decisionmaking body of a VSP. We denote these variables as *ngo_involv* and *public_involv*.[9]

To measure the overall stringency of a program (H3) scholars have focused on the design of standards, monitoring, and enforcement mechanisms (Prakash & Potoski 2007; Potoski & Prakash 2009; Fransen & Burgoon 2011). Following these works, we construct an indicator that comprises three components: a standard's degree of obligation, its scope, and the robustness of its monitoring procedures. Our measure of stringency is constructed as a simple average over variables capturing these three dimensions. Regarding the first dimension, the SMD reports the "degree of obligation" of the individual requirements contained in a standard. Five degrees of obligation are distinguished. In this regard, a requirement can be a recommendation (first type), or implementation can be requested within five years (second), within three years (third), within one year (fourth), or immediately (fifth). We code a numeric version of the degree of obligation by assigning values from 1 to 5 to the above listed typologies. The degree of obligation of the standard (*comp_str*_1) is then measured as the simple average of the degree of obligation across all its requirements. With regard to the scope of a standard, we create a variable (*comp_str*_2) that counts the number of requirements explicitly referenced by a program. Turning to the robustness of monitoring and evaluation procedures, we use information of the type of audits that are required by a program. We define a variable (*comp_str*_3) equal to 0 when only first party auditing is required, taking a value of 1 when second party auditing is required, and equal to 2 when third party auditing is required. The three components (*comp_str*_1; *comp_str*_2; and *comp_str*_3) are normalized between 0 and 1. The simple average across components denoted by *stringency* captures the overall stringency of a program.[10]

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We now turn to the operationalization of our *external* regressors. To examine the effect of "crowding" (H4), we count the number of programs that operate in the same product category (e.g. coffee) and geographical region (e.g. South America). Following assumptions made in organizational ecology, our reasoning is that crowding intensifies the level of competition over material and ideational resources between programs (Hannan & Freeman 1989). We use information about the geographical and product scope of a VSP to create a crowding indicator. It measures the average number of programs that each VSP confronts across the region-product-specific environments in which it operates. Formally, we first compute for each combination of geographic region (r) and product category (p) the total number of programs that operate in region r covering product p. We call those numbers $tot(r, p)$. We then take a simple average of $tot(r, p)$ across those pairs (r, p) to which the particular program s applies. The resulting variable is given by:

$$crowding_s = \frac{\sum_{(r,p)} tot(r,p) \times \mathbb{I}_s(r,p)}{\sum_{(r,p)} \mathbb{I}_s(r,p)} \tag{2}$$

where $\mathbb{I}_s(r,p)$ is an indicator that takes a value of 1 if program s covers product p in region r (and 0 otherwise).

Our second external regressor captures peer pressure (H5). It consists of the average transparency scores of programs that operate in the same product category and geographical region. Similar measures have been used by other scholars to examine peer pressure effects in domestic politics and in the context of IOs (cf. Berliner 2014). Formally, we define our peer pressure indicator as follows:

$$peer_press_s = \frac{\sum_{(r,p)} \left(\frac{\sum_{\sigma \neq s} TI_\sigma \times \mathbb{I}_\sigma(r,p)}{\sum_{\sigma \neq s} \mathbb{I}_\sigma(r,p)} \right) \times \mathbb{I}_s(r,p)}{\sum_{(r,p)} \mathbb{I}_s(r,p)} \tag{3}$$

with $\mathbb{I}_s(r,p)$ as the product-region indicator defined above. In other words, for each pair (r, p) to which a program s applies, we take the average of the transparency index scores of the other programs that are active in the same region r covering product p. We then average the result across all pairs (r, p) to which program s applies.

To examine the effect of metagovernance (H6) we use membership in the focal metagovernance organization in the field of transnational sustainability governance. This is the ISEAL Alliance (Dingwerth & Pattberg 2009; Loconto & Fouilleux 2014). We create a dummy variable labeled *metagov*, taking a value of 1 if the program is either a full or an associate member of ISEAL (0 otherwise).

5.2. Empirical model

We test the hypotheses discussed in the previous section by fitting a linear regression model with the transparency indicators as dependent variables and the six determinants as regressors of interest. The baseline empirical model is given in the following equation:

$$\begin{aligned} TI_s = \alpha + \beta_1\, ngo_involv_s + \beta_2 public_involv_s + \beta_3 stringency_s + \beta_4 crowding_s + \beta_5 peer_press_s \\ + \beta_6 metagov_s + \gamma tech_ctrl_s + u_s \end{aligned} \tag{4}$$

where α is a constant term and u_s the error term. The variable *tech_ctrl* is a technical control that counts the number of areas of operation – that is, decisionmaking, standard setting, verification, and dispute settlement – in which a program is active. In our sample, not all programs perform all functions. This information has been taken into account in the construction of the transparency scores of these programs. This generates mechanical patterns of correlation between *tech_ctrl* and *TI*. Controlling for the number of areas in which a program is active therefore cleans our estimates from potential omitted variable bias.

In addition, we augment the baseline model with a number of controls. The first is the age of a program, *age* (number of years since inception). This controls for temporal dynamics that may affect the transparency of a program. For example, it is conceivable that programs that were created in different time periods faced varying institutional pressures to disclose information. Because of processes of path dependency, these early design decisions may

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Table 2 Summary statistics

Variable	Mean	Median	SD	Min	Max
TI^a	0.232	0.250	0.273	0	1
TI^b	0.333	0.300	0.247	0	1
TI^c	0.401	0.333	0.240	0	1
TI^d	0.449	0.429	0.239	0	1
ngo_involv	0.265	0	0.444	0	1
public_involv	0.133	0	0.341	0	1
stringency	0.694	0.716	0.138	0	0.941
crowding	18.703	20.241	6.246	5.271	31.263
$peer_press^a$	0.201	0.200	0.053	0	0.360
$peer_press^b$	0.317	0.320	0.058	0	0.457
$peer_press^c$	0.395	0.400	0.065	0	0.521
$peer_press^d$	0.451	0.459	0.072	0	0.567
metagov	0.195	0	0.398	0	1
tech_ctrl	3.310	4	0.825	1	4
Age	15.549	14	8.604	2	49
Food_sector	0.690	1	0.464	0	1
Capacity	0.469	0	0.501	0	1

Summary statistics are computed on the estimation sample of 113 observations.

still influence their transparency practices today (Auld 2014; Bloomfield & Schleifer 2017). Similarly, age may be correlated with the involvement of NGOs or public actors as well as with the program's degree of stringency. Second, we control for a VSP's activity within the food sector by means of a dummy variable, *food_sector*, taking a value of 1 if at least part of the production covered by a program is food-related. The inclusion of this variable is meant to control for any variation in both transparency practices and in any regressor of interest that can be explained by the political dynamics specific to the food sector. It has been argued that the reputational stakes of firms in the food sector tend to be higher and that this may exert upward pressure on best practice compliance, including transparency practices (van der Ven 2015, p. 7). Finally, we create a variable *capacity* to control for the organizational capacity of a program. The rationale is that disclosing information generates costs and that programs with high organizational capacity are in a better position to absorb these costs (Mitchell 1998; Grigorescu 2007). In addition, organizational capacity may be correlated with a program's ability to comply with the standards of a meta-governance organization. As a proxy for high organizational capacity, we identify those programs in our sample that, next to their headquarters, operate local offices. The dummy takes a value of 1 if the program operates local offices (0 if not).[11] Table 2 provides summary statistics computed on the estimation sample (n = 113) for the variables used in the baseline analysis.

6. Results

We conduct two sets of ordinary least squares (OLS) estimations; the results are reported in Table 3. First, we run four regressions – one for each version of the transparency index – where we include the six explanatory variables of interest plus *tech_ctrl* (models 1–4 of Table 3). Second, we re-estimate these four regressions, including the three controls defined above (models 5–8).

How do the estimates in Table 3 speak to our theoretical hypotheses? Let us start from the discussion of the internal determinants of transparency. First, our variable *ngo_involv* does not emerge as a significant predictor of greater transparency. Point estimates for the *ngo_involv* coefficient are always positive but never statistically significant, suggesting that H1 is not confirmed by our data. Second, in support of H2, the involvement of public actors is associated with higher levels of transparency. The estimates of the *public_involv* coefficient are positive and statistically significant across all regressions. Moreover, the magnitude of the estimated coefficients is remarkably stable. In this regard, the point estimate in model 2 implies that, ceteris paribus, involvement of a public actor in a VSP's central decisionmaking body is associated with an increase in TI^b of 0.128, which corresponds to

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Table 3 Estimation results

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Dep var:	TI^a	TI^b	TI^c	TI^d	TI^a	TI^b	TI^c	TI^d
ngo_involv	0.050	0.029	0.013	0.003	0.048	0.028	0.014	0.005
	(0.068)	(0.060)	(0.057)	(0.055)	(0.072)	(0.063)	(0.059)	(0.058)
public_involv	0.136*	0.128**	0.126**	0.125**	0.140*	0.135**	0.134**	0.134**
	(0.070)	(0.063)	(0.061)	(0.061)	(0.071)	(0.066)	(0.065)	(0.066)
stringency	0.315	0.278	0.240	0.209	0.348*	0.294	0.247	0.211
	(0.198)	(0.187)	(0.186)	(0.190)	(0.194)	(0.180)	(0.179)	(0.184)
crowding	−0.010***	−0.009**	−0.008**	−0.007**	−0.009*	−0.008*	−0.008*	−0.007*
	(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
peer_press	−0.436	−0.383	−0.286	−0.218	−0.473	−0.404	−0.311	−0.250
	(0.441)	(0.333)	(0.303)	(0.300)	(0.457)	(0.344)	(0.318)	(0.319)
metagov	0.213***	0.196***	0.188***	0.182***	0.210***	0.192***	0.181***	0.175***
	(0.073)	(0.065)	(0.060)	(0.057)	(0.073)	(0.065)	(0.060)	(0.057)
Controls					✓	✓	✓	✓
Observations	113	113	113	113	113	113	113	113
R-squared	0.278	0.293	0.286	0.271	0.281	0.296	0.291	0.278

*$P < 0.10$, **$P < 0.05$, ***$P < 0.01$. All specifications include a technical control (*tech_ctrl*), measuring the number of areas in which a program is active. The controls included in the last three columns are: (i) *age*, (ii) *food_sector*, and (iii) *capacity*. The variable *peer_prs* is consistent with the specific version of the transparency index in each column. Robust standard errors are reported between brackets.

almost 52 percent of a standard deviation for TI^b. This can be interpreted as a relevant variation in transparency as it corresponds to more than half of the average variation in transparency from the sample mean. Finally, the estimated coefficients for *stringency* display a positive sign but are never statistically significant. Therefore H3, although not discarded by the estimation results, is not robustly confirmed.

Turning to the external determinants of transparency, the empirical analysis provides a number of very robust findings. First, crowding (H4) is a significant predictor of transparency. More precisely, crowding is associated with less transparency. Estimates for the relevant coefficient are robust in terms of sign across all specifications, they are highly significant from a statistical point of view in the four models without controls, and they largely meet the 10 percent requirement in terms of statistical significance when additional controls are added to the specification.[12] Interestingly, the magnitude of the point estimates seem to be attenuating with higher weight placed on the shallow level of disclosure practices in the construction of the transparency index. As an illustration of the relationship implied by our estimates, consider again model 2. The estimated *crowding* coefficient implies that, all other things being equal, one standard deviation increase in *crowding* is associated with a decrease in TI^b by almost a quarter (23 percent) of a standard deviation. As for precision, if one replicates the same quantification with the endpoints of a 90 percent confidence interval around the estimated coefficient, the decrease in TI^b associated with a one standard deviation increase in *crowding* amounts to 38 percent and 8 percent of a standard deviation for the left and the right endpoint, respectively. Second, hypothesis H6 on the positive link between metagovernance and transparency is strongly confirmed by our data. Estimated coefficients for *metagov* are positive and very precisely estimated across all proposed empirical models. Their magnitude is also remarkably robust. The point estimate reported in model 2 and its 90 percent confidence interval imply that, ceteris paribus, being a member of ISEAL would be associated with a TI^b score of 0.196 units higher (almost 80 percent of a standard deviation), plus/minus 0.107 units. Third, the estimated coefficient of the variable *peer_press* remains statistically non-different from zero across all specifications, suggesting the lack of a systematic relationship between peer pressure and transparency practices in our data. Finally, let us briefly report on the estimated coefficients for the controls.[13] The point estimates of the variable *tech_ctrl*'s coefficients are all positive, statistically significant, very precisely estimated, and with an average value of 0.075 across the eight models. On the contrary, none of the estimated coefficients for the three controls *age*, *food_sector*, and *capacity* turns out to be statistically different from zero.

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Table 4 Bivariate probit regression estimates

	Model 1	Model 2	Model 3	Model 4
Area of operations	Decisionmaking	Standard setting	Verification	Dispute settlement
public_involv	0.868**	1.298***	0.123	−0.300
	(0.409)	(0.411)	(0.413)	(0.589)
crowding	−0.027	−0.064***	−0.025	0.018
	(0.028)	(0.023)	(0.031)	(0.045)
metagov	0.972***	0.970***	0.726**	0.700*
	(0.355)	(0.328)	(0.360)	(0.388)
Observations	113	107	87	67
Pseudo R-squared	0.152	0.231	0.092	0.201

*$P < 0.10$, **$P < 0.05$, ***$P < 0.01$. Robust standard errors are reported between brackets.

6.1. Further estimations

Here we address potential concerns regarding our baseline empirical exercise presented above. First, we replicate the estimations of models 1–4 in Table 3 by removing the only two VSPs that score the maximum value (1) across all transparency indicators from the sample. The estimates show high stability of all of our baseline findings with respect to these outliers. The results do not change when including the three additional controls *age*, *food_sector*, and *capacity*. Moreover, baseline results remain robust when augmenting the specification with an additional control capturing the democratic quality of domestic institutions.[14] The reasoning is that the institutionalization of democratic norms in a VSP's country of origin could influence transparency practices. Furthermore, we use an alternative control to capture sector specific effects. To this end, we replace the *food_sector* variable with a dummy that takes a value of 1 if a program operates in more than one industry sector (0 if the program operates in a single sector).[15]

Finally, we investigate whether the patterns that come out as statistically significant for the aggregate transparency indexes change when considering disclosure practices in each individual area of operations, that is, decisionmaking, standard setting, verification, and dispute settlement. In doing so, we focus on deep levels of transparency only. Formally, we run a bivariate probit regression for each dummy variable $t_{s;\ c,\ deep}$, taking a value of 1 if the program *s* publicly discloses information at a deep level on the corresponding area of operation *c*. The estimation sample will change across regressions, reflecting the fact that not every VSP is active in all areas of operations. We estimate a parsimonious specification featuring only the three regressors whose coefficients were statistically significant in the baseline model. These are *public_involv*, *crowding*, and *metagov*. The results do not change when estimating the model with the full set of explanatory variables. Estimates for the four probit models are reported in Table 4.[16]

The results support the findings derived from the baseline exercise. The estimated coefficients for *public_involv* have a positive sign when statistically significant, suggesting a positive relationship between the involvement of public actors and information disclosure in the areas of decisionmaking (model 1) and standard setting (model 2). The estimated coefficient for *crowding* has a negative sign in the three models assessing transparency in decisionmaking, standard setting, and verification. Statistical significance is very high in the case of standard setting. Lastly, *metagov* is a significant determinant of transparency across all individual areas of operations.

7. Discussion and conclusion

The spread of "new" transnational governance has prompted questions about its accountability. Without recourse to principal–agent forms of accountability, scholars and practitioners have proposed transparency as a potential solution. Yet as we have shown, very few studies have documented and explained variation in the practice of transparency in transnational governance. Our analysis contributes to filling this gap by examining internal and external factors associated with deep (shallow) transparency practices in a group of 113 voluntary programs focused on sustainability. Several findings and implications follow. First, with internal determinants, our analysis underscores the importance of "who governs" (van der Ven 2015, pp. 5–6). We found a robust positive

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relationship between the involvement of public actors (e.g. representatives from state agencies and ministries) and information disclosure, corroborating one of van der Ven's (2015) key findings in this area.[17] This result directly speaks to ongoing debates about the interactions between "new" transnational governance and "old" state-led governance (Abbott & Snidal 2009a; Eberlein *et al.* 2014; Overdevest & Zeitlin 2014). It has been argued that greater involvement by government actors in transnational governance could promote "substantive principles and procedures derived from public law to reinforce transparency and accountability" (Abbott & Snidal 2009b, p. 558). With a focus on transnational sustainability governors, our results lend empirical support to such claims. We find that, everything else being equal, VSPs with direct public involvement have "deeper" transparency regimes than those that do not, and that this relationship appears to derive from public actors' participation in decisionmaking. As argued throughout this article, deep transparency is a necessary condition to enable information users (e.g. critical NGOs, consumer groups, and investors) to act as accounting actors in transnational governance.

Interestingly, NGO involvement does not have the same effect. Our data does not point to a statistically significant relationship between NGO involvement and greater transparency (Grigorescu 2007; Abbott & Snidal 2009a; van der Ven 2015). Various explanations are possible for the absence of an "NGO effect." One is that "insider" NGOs may have little incentives to make sensitive information available to critical "outsider" NGOs or other stakeholders, a possibility consistent with the findings of Auld and Gulbrandsen's (2010) comparison of the Marine Stewardship Council and FSC. This explanation would also be consistent with known disagreements among NGOs over the value of market-based sustainability governance (e.g. Schleifer 2019). An alternative explanation is that NGOs, even if formally involved in the central governance body, may lack the institutional power to push through their positions. These are plausible arguments meriting further research that can assess the capacity and institutional power of participating NGOs, as well as whether insider-outsider dynamics are at play in shaping the strategic value of different levels of transparency across a VSP's activities (Gulbrandsen & Auld 2016).

While we do not find statistical support for arguments linking direct NGO participation to transparency, the analysis revealed a very robust relationship between civil society-led metagovernance and information disclosure. Several qualitative studies have shown how metagovernance organizations like the ISEAL Alliance play an important role as norm entrepreneurs in transnational sustainability governance (Dingwerth & Pattberg 2009; Loconto & Fouilleux 2014). Our analysis confirms a positive correlation between ISEAL membership and high levels of transparency, suggesting that this specific NGO – working as a metagovernance organization – has been able to hold members to account against norms of procedural best practices relating to decisionmaking, standard setting, verification, and dispute resolutions. Indeed, compared to the public actor effect, the disaggregate probit model found statistically significant relationships between ISEAL membership and all four areas of operations (decisionmaking, standard setting, verification, and dispute resolution). However, this result needs to be interpreted with care. Based on the statistical analysis alone, it is not possible to clearly establish the direction of the effect. In this regard, the pattern we observe could also partly be the consequence of self-selection – that is, the fact that more transparent VSPs worked to establish or joined ISEAL in order to differentiate themselves from their less transparent peers.

Our external determinants reveal two further and notable findings. First, there is no statistical association between our measure of normative peer pressure and information disclosure. A possible explanation for this lack for convergence could be of a political–institutional nature. Examining the non-convergence of VSPs in the field of transnational labor governance, Fransen (2011) argued that this is a result of the persistence of political differences between interest groups creating and supporting these arrangements. A similar explanation could be behind the lack of normative peer pressure dynamics in the wider population of VSPs. Second, our analysis reveals a robust negative relationship between crowding and transparency. Drawing on the literature on organizational ecology (Hannan & Freeman 1989), we assumed that crowding intensifies resource competition between VSPs. Our results show that, as these pressures intensify, programs are less willing to openly share sensitive information with their competitors (and the general public). Interestingly, the effect of crowding is strongest in Model 1, which focuses on deep transparency only.[18] However, at least one alternative interpretation of the relationship between crowding and transparency merits discussion. In this regard, it is possible that very high (or low) transparency levels of first-mover programs motivate other actors in these fields to create competitor programs that better

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correspond to their preferences (see Fransen & Conzelmann 2015). For example, more "conservative" business actors may decide to create their own programs in response to an overly revelatory first-mover NGO-led program. In other words, as in the case of metagovernance, some of the causality may run the other way around.

Overall, these factors begin to explain differences in the information disclosure practices of VSPs. In cases in which information users, such as critical NGOs, consumer groups, and investors have open access to meeting minutes, audit reports, and dispute settlement decisions, we expect a higher likelihood that market pressures, public scrutiny, and self-reflection foster accountability in transnational governance. Clearly, "shallow transparency" will not do the job. At the same time, one should not underestimate the difficulties of "embedding" the provided information in the behavior of potential accounting actors (Fung *et al.* 2004, pp. 10–15). In this regard, information that is provided but not used will not promote accountability either. But this is a second-order question, and our findings suggest that the causal chain gets interrupted at an earlier stage. In the field of sustainability politics, the vast majority of transnational governors simply disclose too little information to make the transparency action cycle swing (Fung *et al.* 2008).

Beyond accountability, the dominance of shallow transparency in our dataset points to the need for further analysis of the strategic value of privacy versus transparency from the perspective of VSPs. Operating in markets, these sustainability governors confront longstanding norms of commercial secrecy where certain information is closely guarded for competitive reasons. Moreover, with few exceptions, work on VSPs is skewed toward certain cases (e.g. the forest and coffee sectors) or particular programs (e.g. the FSC) and particular metagovernors (e.g. ISEAL Alliance). Indeed, transparency also affects the ease with which academics can study different VSPs, creating a potential bias in our knowledge toward those programs that disclose more information about their operations. Incomplete and variegated transparency in the practices of transnational governance has, in other words, significance beyond questions of accountability.

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Notes

1 See Auld and Gulbrandsen (2010) for a notable exception.

2 The quantitative research agenda on transnational sustainability governance is still in its infancy. See Marx and Cuypers (2010), van der Ven (2015), and Darnall *et al.* (2017) for important exceptions.

3 More details can be found under www.standardsmap.org. Please note that the online tool does not contain the full SMD, which was used as a basis for this article. [Last accessed 7 February 2019.]

4 Another important database of VSPs is the Ecolabel Index (http://www.ecolabelindex.com/). It includes a larger number of VSPs than the SMD (in particular more domestic programs and corporate codes of conduct). However, it contains fewer data points per program – approximately 60 in comparison to approximately 1,000 data points per VSP in the full SMD. [Last accessed 7 February 2019.]

5 A description of the ITC's data collection protocols can be found at: http://www.intracen.org/itc/market-data/standards-map/participating-standards/ [Last accessed 7 February 2019.]

6 For a more detailed introduction to the SMD please see Fiorini et al. (2018).

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7 Please note that normalizing the transparency indexes to vary between 0 and 1 makes the actual levels of the weights irrelevant, as long as the ratio between them is kept constant. For instance, any other weighting scheme that assigns a weight to the shallow level of transparency that is half of that assigned to the deep level would generate the same transparency index as TI^c.

8 See https://www.isealalliance.org/our-members for details. [Last accessed 7 February 2019.]

9 Unfortunately, our data does not allow us to distinguish between public actors from democracies and non-democracies (Grigorescu 2007). Therefore, we can only examine the general effect of public actor involvement on transparency.

10 The main results of the baseline analysis conducted do not change if we take each individual component of our composite stringency indicator (degree of obligation, standard scope, and audit type) as an empirical measure of stringency.

11 The baseline results presented remain robust to an alternative proxy for organizational capacity which takes a value of 0 when the program has no local office beyond the headquarter, 1 when the system has one local office, 2 for two to four local offices, 3 for five to nine local offices, and 4 for 10 or more local offices.

12 The p-values associated to the estimates of the *crowding* coefficient in models 4 to 8 in Table 3 are equal to 0.055, 0.053, 0.065, and 0.083, respectively.

13 For the sake of space, these are not listed in Table 3.

14 We measure domestic institutions by using the Voice and Accountability indicator from the World Bank's Worldwide Governance Indicators corresponding to the country where a VSP has its headquarters (World Bank 2016).

15 Results of these robustness exercises are not reported in the paper because of space considerations but are available upon request.

16 Our probit models are based on the following assumptions: (i) for each area of operations, c, there exists an underlying unobservable (latent) variable capturing VSPs' utility from disclosing information on that particular area (U^c); (ii) this utility is a linear function of the explanatory variables included in the specification. Formally:

$$U_s^c = \alpha + \beta_1\, public_involv_s + \beta_2 crowding_s + \beta_3 metagov_s + u_s$$

with u being an error term distributed normally with mean 0; and (iii) when U_s^c is > 0 we observe $t_{s;\ c,\ deep} = 1$, when instead the utility is negative, we observe $t_{s;\ c,\ deep} = 0$. The estimates reported in Table 4 should be interpreted as the estimated marginal effect of each regressor with respect to the latent variable U^c. Marginal effects on the probability of observing $t_{s;\ c,\ deep} = 1$ (which have the same signs as the estimated marginal effects in Table 4) are available upon request.

17 Unfortunately, our data does not allow us to distinguish between government actors from democratic and non-democratic countries. However, following the logic of the argument laid out above, we hypothesize that the effect should only hold for state actors from democratic countries.

18 van der Ven found no statistically significant effect between crowding and best practice compliance, although the sign of his competition coefficient is likewise negative. A possible explanation for such discrepancies is the differences in the samples between the two studies. For example, by not including multi-sector programs in his analysis, van der Ven may have underestimated the effect of crowding – a caveat he discusses himself (van der Ven 2015, p. 14).

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